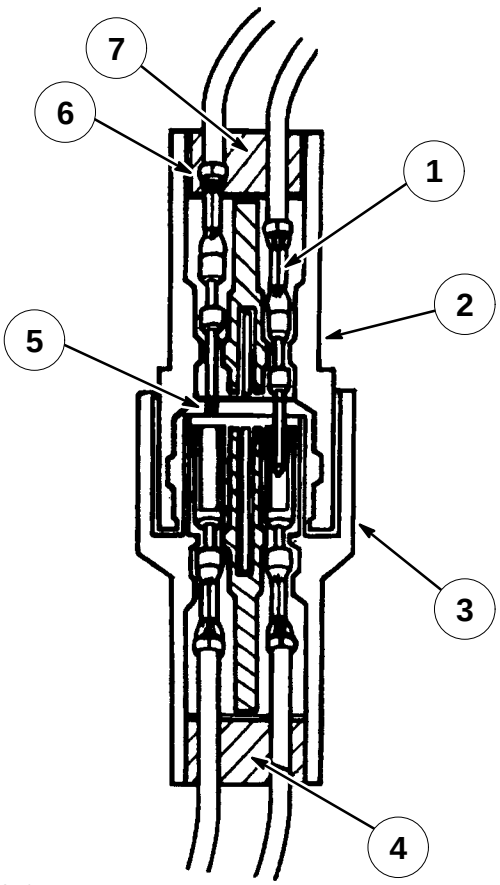
Troubleshooting wiring harness and connector hidden concerns **-1**

The following illustrations are known examples of wiring harness, splices and connectors that will create intermittent electrical concerns. The concerns are hidden and can only be discovered by a physical evaluation as shown in each illustration.

NOTE: Several components, such as the PCM, utilize gold plated terminals in their connections to the wiring harness. If those terminals need to be replaced, they must be replaced with a gold plated terminal.

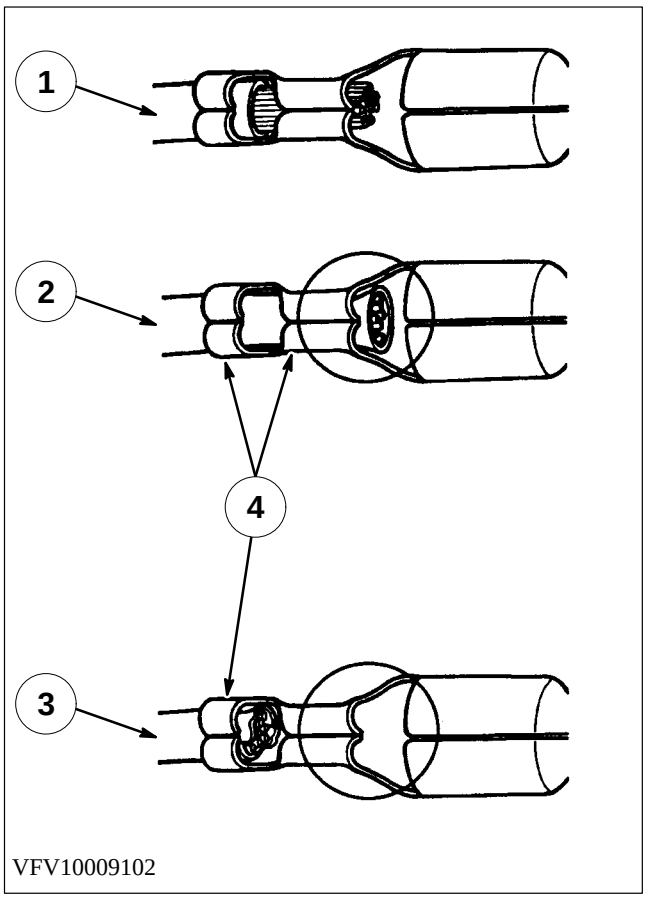


VFV10009101

Terminal not properly seated

1 = Locked terminal
2 = Male half
3 = Female half
4 = Seal
5 = Intermittent contact
6 = Unlocked terminal (Hidden by wire seal)
7 = Seal

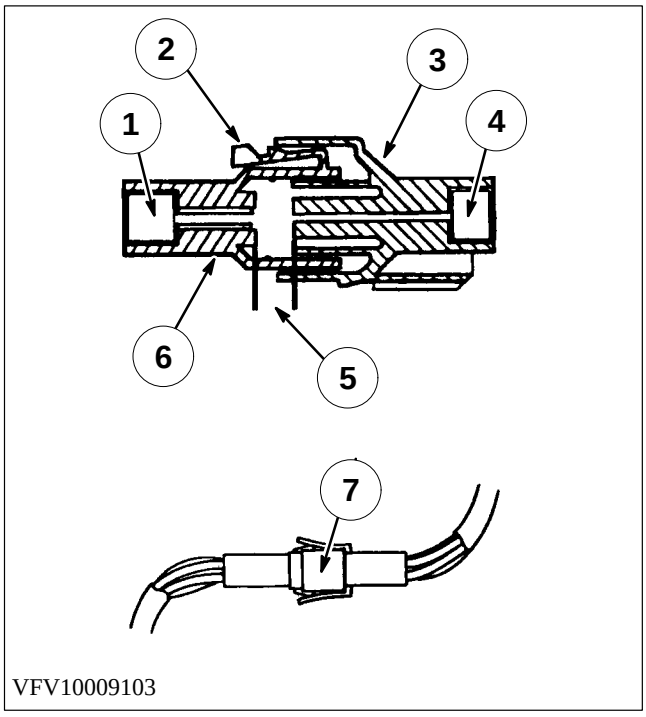
Check for unlocked terminals by pulling each wire at the end of the connector.



VFV10009102

Defective insulation stripping

1 = Proper crimp
2 = Insulation not removed
3 = Wire strands missing
4 = Intermittent signals through pierced insulation

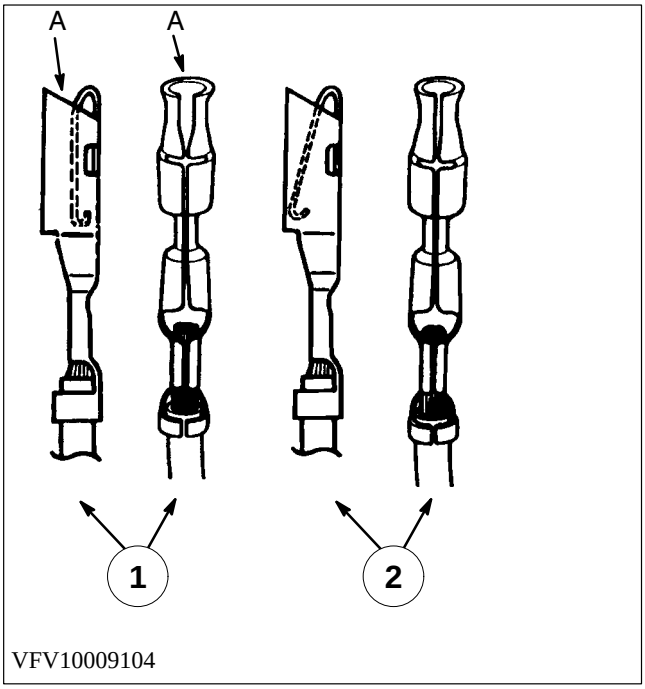


VFV10009103

Partially mated connectors

1 = Seal
2 = Displaced tab
3 = Female half
4 = Seal
5 = Intermittent contact
6 = Male half
7 = Intermittent contact

Lock may be displaced into an unlocked position; pull on the connector to verify the lock.

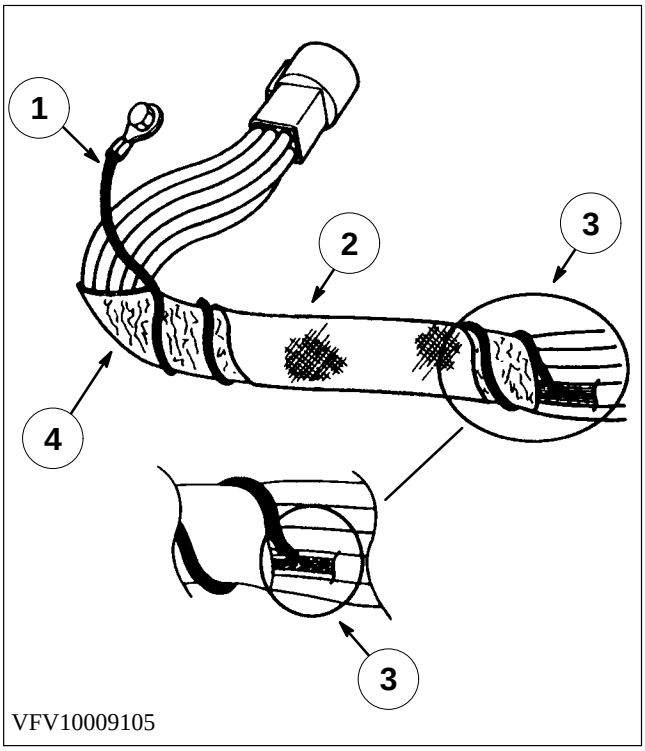


VFV10009104

Deformed (enlarged) female terminals

1 = Enlarged
2 = Normal

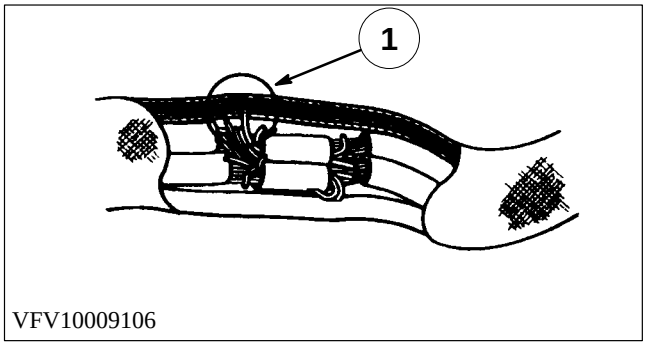
Any probe entering the terminal may enlarge the contact spring opening creating an intermittent signal. Insert the correct mating terminal (Location A) from the service kit and feel for a loose fit.



VFV10009105

Electrical short inside the harness

1 = Solder coated wire to ground
2 = Harness protective tape
3 = Intermittent short
Solder coated wire pierced through the insulation of another circuit
4 = Grounding foil

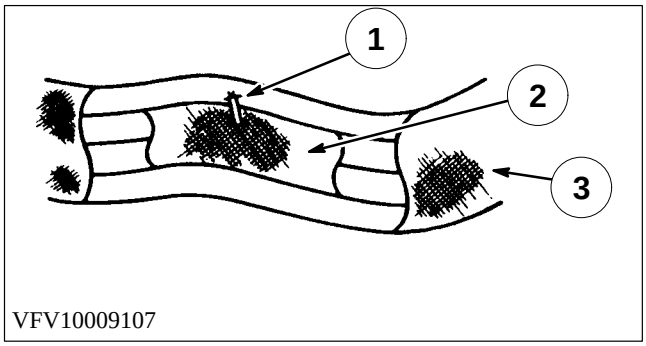


VFV10009106

Electrical short within the harness

Splice tape removed

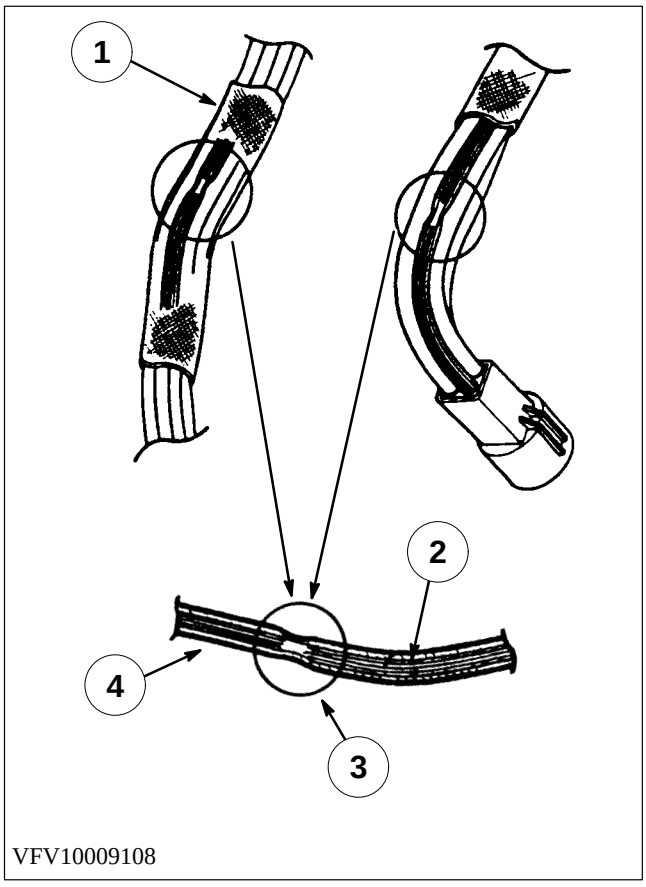
1 = Intermittent short



VFV10009107

Splice covered

1 = Wire strand
2 = Splice tape
3 = Harness tape



VFV10009108

Broken wire strands in harness

1 = Wiring harness tape
2 = Wiring strand
3 = Broken strands intermittent signal
4 = Circuit insulation

Remove the tape and flex/feel each circuit for a reduction in diameter at break.